CIQA - Analytical Quantum Error Correction

The ‘1:5 QEC’ by **arcadia**lab**.fr**

- Extra quick tech brief -

Every existing Quantum Error Correction code implementation either hasn't left simulation, requires hardware that doesn't exist yet, is memory-only, or can't do full Pauli correction at computational depth.

CIQA is the only one that can be used today on real hardware for actual computation. It features an exact 1:5 encoding ratio, requiring 5 physical data qubits per logical qubit.

CIQA is not a stabilizer code.
There is no notion of weight, no minimum-distance decoding, and no threshold that scales with block size in the stabilizer sense.

Table 1 - All overhead ratios are based on data qubits only - Table based on published papers, **see references page 4**

QUANTUM
ERROR CORRECTION STATE
2026

	ARCADIALAB.FR	IBM	Google	QUANTINUUM Microsoft	QUANTINUUM
TECH.	CIQA	BB LDPC	Surface Code d=7	Logical Qubit Virt.	Iceberg d=4
Overhead ratio (logical:physical)	1:5	1:12	1:49	1:7	1:1.67
QuDit handling (dimensions above qubit)	YES	NO	NO	NO	NO
Hardware agnostic	YES	NO	NO	NO	NO
Active computation (not memory only)	YES	NO	NO	NO	YES
Full Pauli correction (at computational depth)	YES	NO	NO	YES	NO
Zero free parameters (fully analytic)	YES	NO	NO	NO	NO
Validated on hardware	YES	NO	YES	YES	YES

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Quick highlight

✅ Overhead ratio

Why the **1:5** ratio is the proper one ?

- Above it, there is nothing new to protect against.
- Below it, unless specific hardware, the correction doesn't work on all axes, or at sufficient depth.

✅ Hardware-agnostic

CIQA is designed for no specific architecture, and produces reliable, consistent results at any depth.

✅ **Active computation** and **full Pauli correction at computational depth** are the minimum requirements to meet for a QEC aiming to be production-ready. **CIQA** is currently the only one to meet both, and more.

Validation

Standalone benchmark - IBM Heron r2, April 3, 2026.

Three independent runs, 8192 shots each, pinned layout, live calibration data.

Circuit	Mean fidelity	SD
Bare qubit, no error	**0,994**	**0,002**
Bare qubit, X error, no correction	**0,015**	**0,002**
CIQA encoded, no error	**0,818**	**0,011**
CIQA, X error, corrected	**0,847**	**0,008**
Mean improvement over unprotected	**+0,832**	**0,008**

IBM hardware was chosen by default as they were the only public, freely accessible QPUs at the time of the experiment. **CIQA**'s encoding works on a topology it was not designed for, with no calibration.

Hayden-Preskill experiment, IBM Heron r2

The Hayden-Preskill circuit models black hole evaporation as a quantum system across multiple entangled registers over several steps. The circuit accumulates decoherence faster than it produces signal, which is why it has never been run past a critical threshold that would allow meaningful measurements.
CIQA prevents the decoherence to happen early on, allowing the very first observation of the black hole evaporation past the Page time.
The experiment ran across three backends, multiple system sizes, and has been reproduced a dozen times. Full dataset, including IBM QPU side, published alongside the paper.

Good to know

Data Overhead: CIQA is based on the [[5,1,3]] perfect quantum code layout, requiring exactly **5 physical** data qudits **per logical** block (the distance parameter doesn't apply to **CIQA**).

Syndrome Extraction: Unlike conventional codes that require a high volume of stabilizer measurements, **CIQA** leverages native qudit dimensions to perform full Pauli correction using only **2 dedicated ancilla qudits** per logical block (one per channel).

Pipeline Automation: When paired with the **CIQS** compiler (*see the **CIQS** tech brief*), the pipeline automatically isolates the **CIQA** encoding prefix, routes the circuit across the topology (like the heavy-hex layout), and applies its non-heuristic optimization **only** to the operational gates outside of that protected prefix.

Links

Compiled source code (free for non-commercial use)
https://github.com/GamesOfArcadia/ciqs

CIQA paper (architecture, derivation, hardware validation):
https://zenodo.org/records/19405503

Hayden-Preskill paper (deep circuit run, full dataset):
https://zenodo.org/records/21326186

References

- **IBM BB LDPC** - DOI: **10.1038/s41586-024-07107-7**
- **Quantinuum Iceberg** - DOI: **10.48550/arXiv.2602.22211**
- **MS+Quantinuum** - DOI: **10.48550/arXiv.2404.02280**
- **Google Willow** - DOI: **10.1038/s41586-024-08449-y**